

BDA400 Assignment 2

Technical Analysis Using R – Preliminary Stage

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Course: BDA400

GitHub Repository:

Technical Analysis – Assignment2

<https://github.com/Parv2003/TechnicalAnalysis/tree/main/Assignment2>

1. GitHub Repository Setup

The TechnicalAnalysis GitHub repository was created for the BDA400 technical-analysis work. The repository uses the main branch and contains the Assignment2 folder.

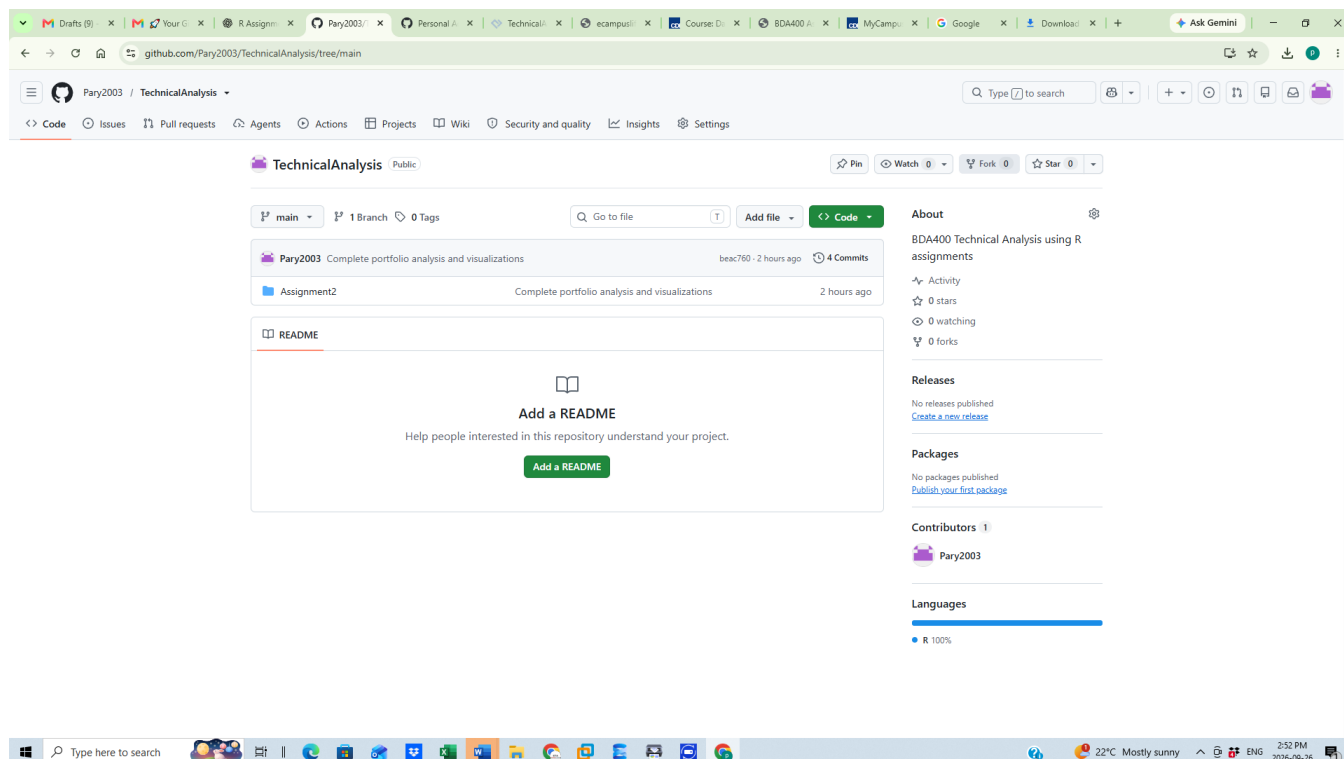


Figure 1. TechnicalAnalysis repository on GitHub showing the main branch and Assignment2 folder.

2. Portfolio File

The portfolio.txt file contains five selected stock symbols: AAPL, MSFT, GOOG, AMZN, and TSLA. These symbols are read by the R script for portfolio-wide processing.

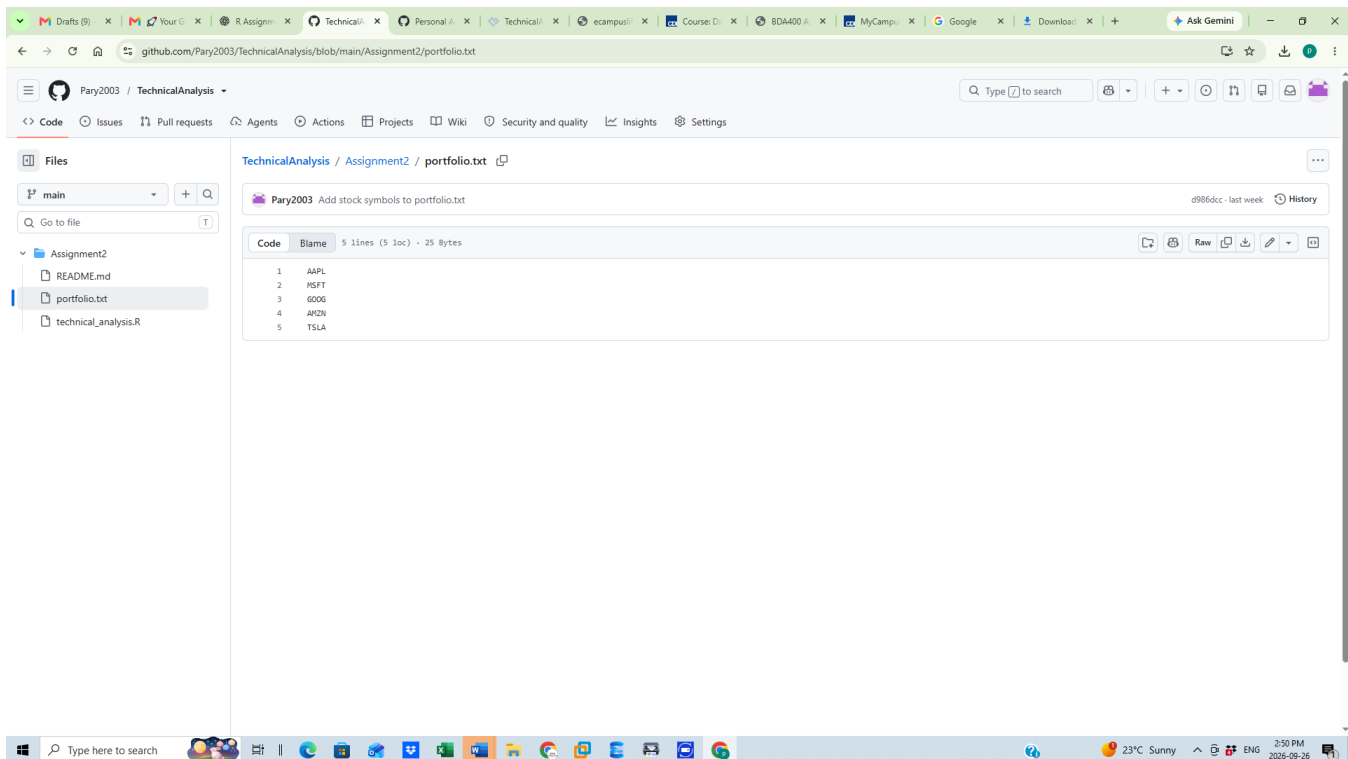


Figure 2. *portfolio.txt* containing the five selected stock symbols.

3. R Packages, Data Import, and Statistics

The R script loads `quantmod` and `TTR`. The `import_stock_data()` function retrieves historical market data from Yahoo, while `calculate_statistics()` begins the calculation of the 20-day simple moving average and descriptive statistics.

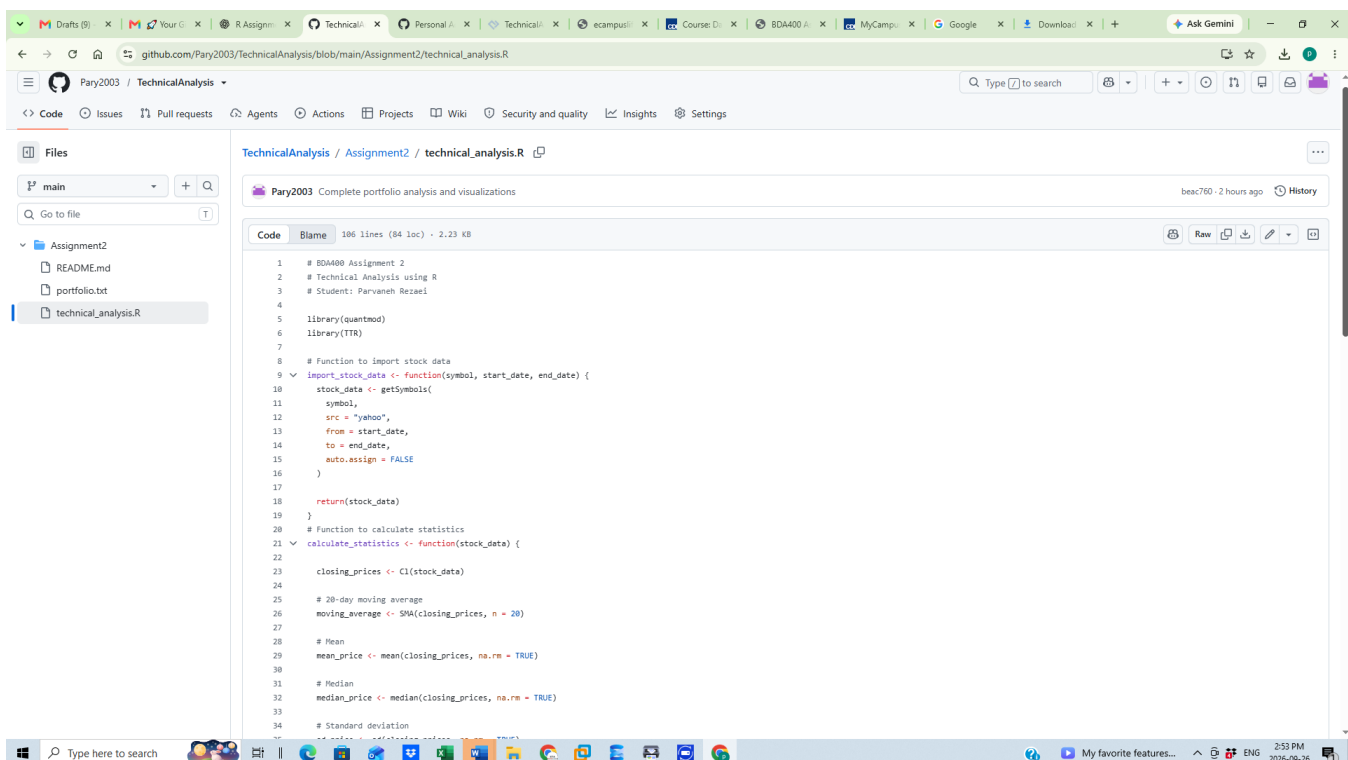


Figure 3. R packages, stock-data import function, and beginning of the statistics function.

The statistics function calculates standard deviation and mode and returns the moving average, mean, median, mode, and standard deviation. The script then imports AAPL data, displays the data, calculates statistics, and creates stock-price visualizations.

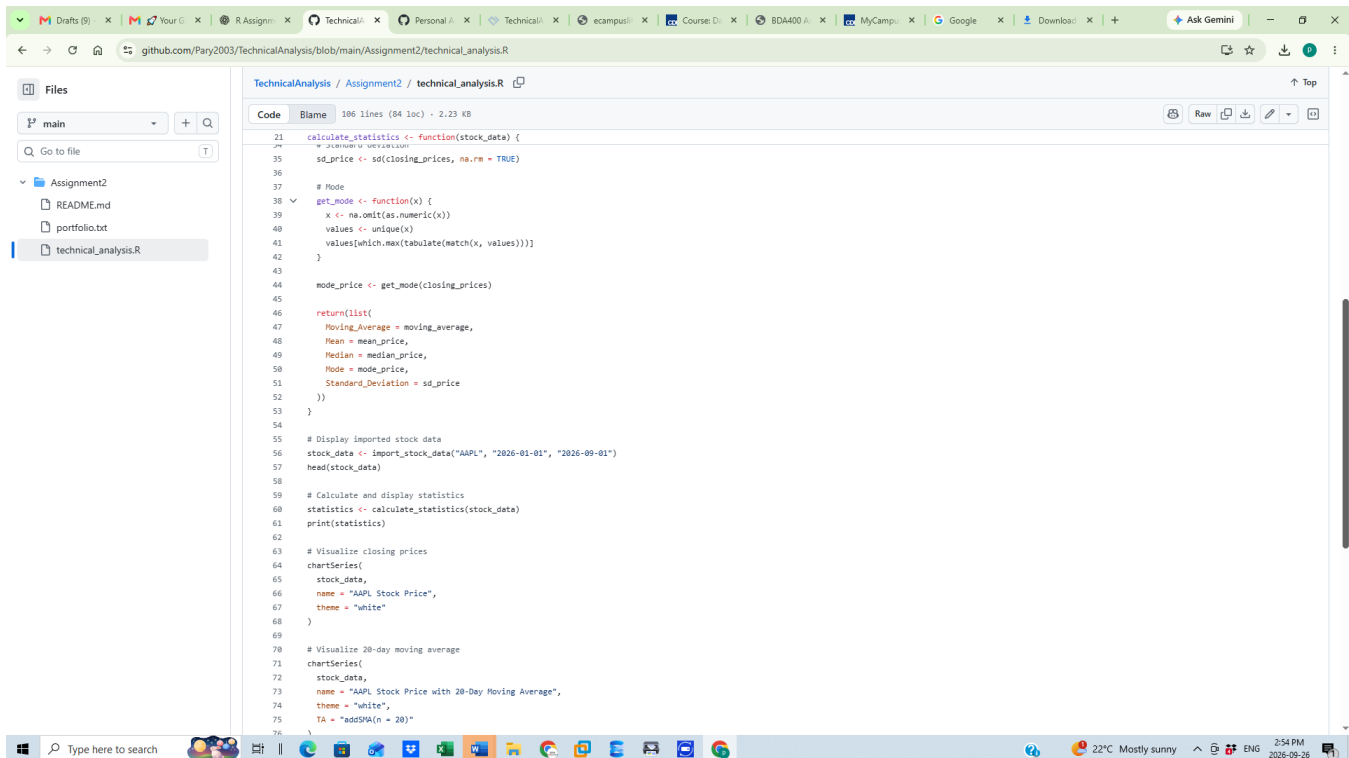
The image is a screenshot of a web browser displaying a GitHub repository. The browser's address bar shows the URL 'github.com/Pary2003/TechnicalAnalysis/blob/main/Assignment2/technical_analysis.R'. On the left side, there is a file explorer showing the repository structure: 'main' branch, 'Assignment2' folder, and files 'README.md', 'portfolio.txt', and 'technical_analysis.R'. The main content area displays the R script 'technical_analysis.R' with line numbers 21 to 76. The script defines a 'calculate_statistics' function that takes 'stock_data' as input and returns a list containing 'moving_average', 'mean_price', 'median_price', 'mode_price', and 'standard_deviation'. It also includes code to import AAPL stock data, display it, calculate statistics, and create two charts: 'AAPL Stock Price' and 'AAPL Stock Price with 20-Day Moving Average'. The Windows taskbar at the bottom shows the time as 2:54 PM on 2026-09-26, with a weather widget indicating 22°C and 'Mostly sunny'.

Figure 4. Completion of statistical calculations and AAPL analysis/visualization code.

4. Portfolio Processing

The script reads the symbols from Assignment2/portfolio.txt and loops through every stock. For each symbol it imports the data, prints a sample, calculates statistics, and generates a chart with a 20-day moving average.

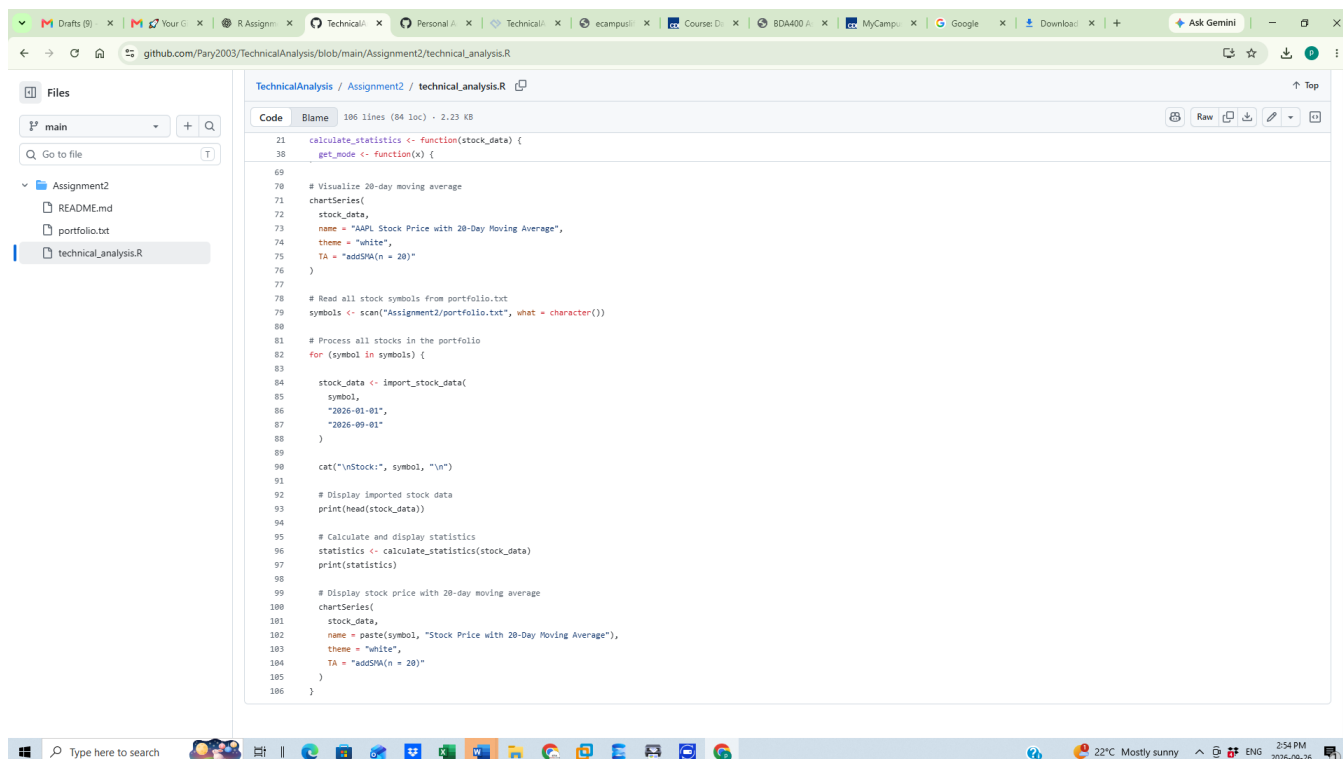


Figure 5. Portfolio loop processing all stock symbols and generating 20-day moving-average charts.

5. Analysis Output

The completed script was executed in Posit Cloud. The final displayed TSLA result shows the stock-price chart with a 20-day moving average together with the calculated statistics in the Console.

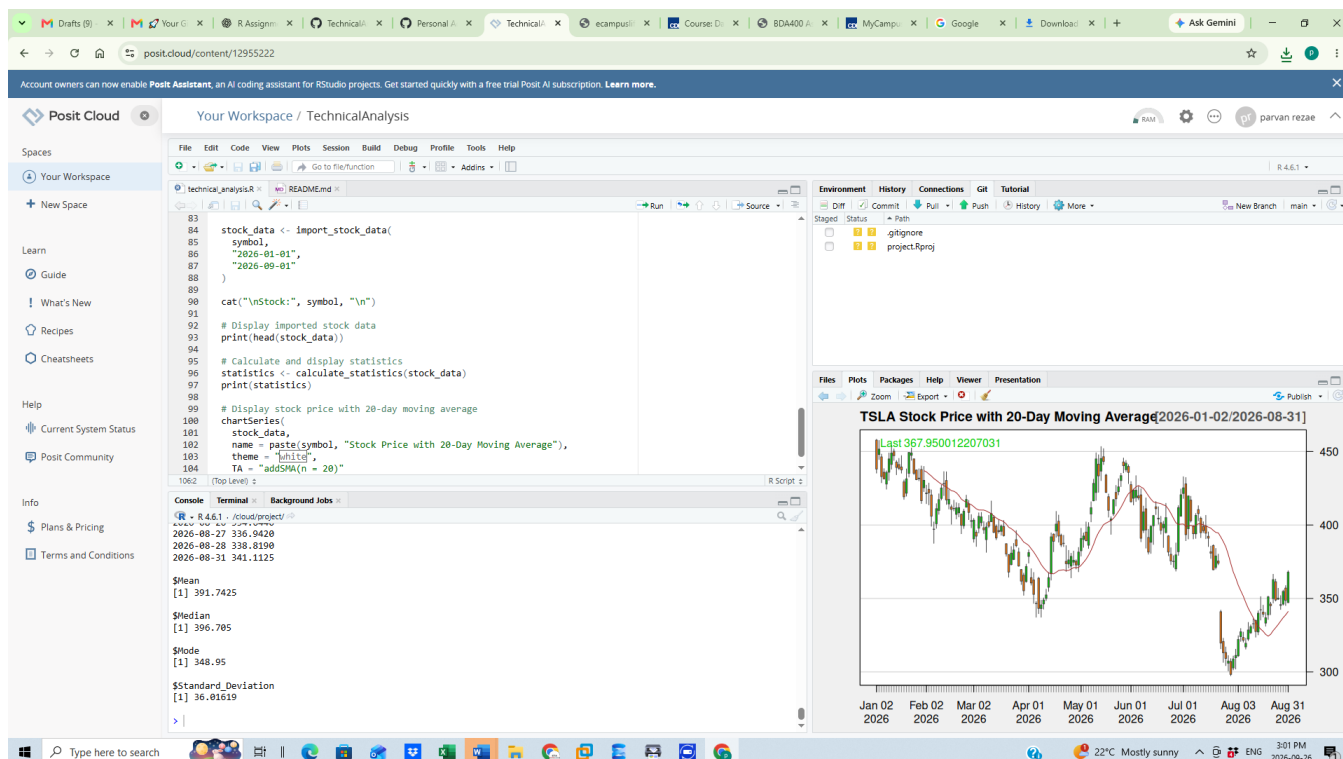


Figure 6. Posit Cloud output showing TSLA statistics and the 20-day moving-average chart.

6. Git and GitHub Synchronization

After completing the analysis, the changes were committed and pushed from Posit Cloud to the GitHub main branch.

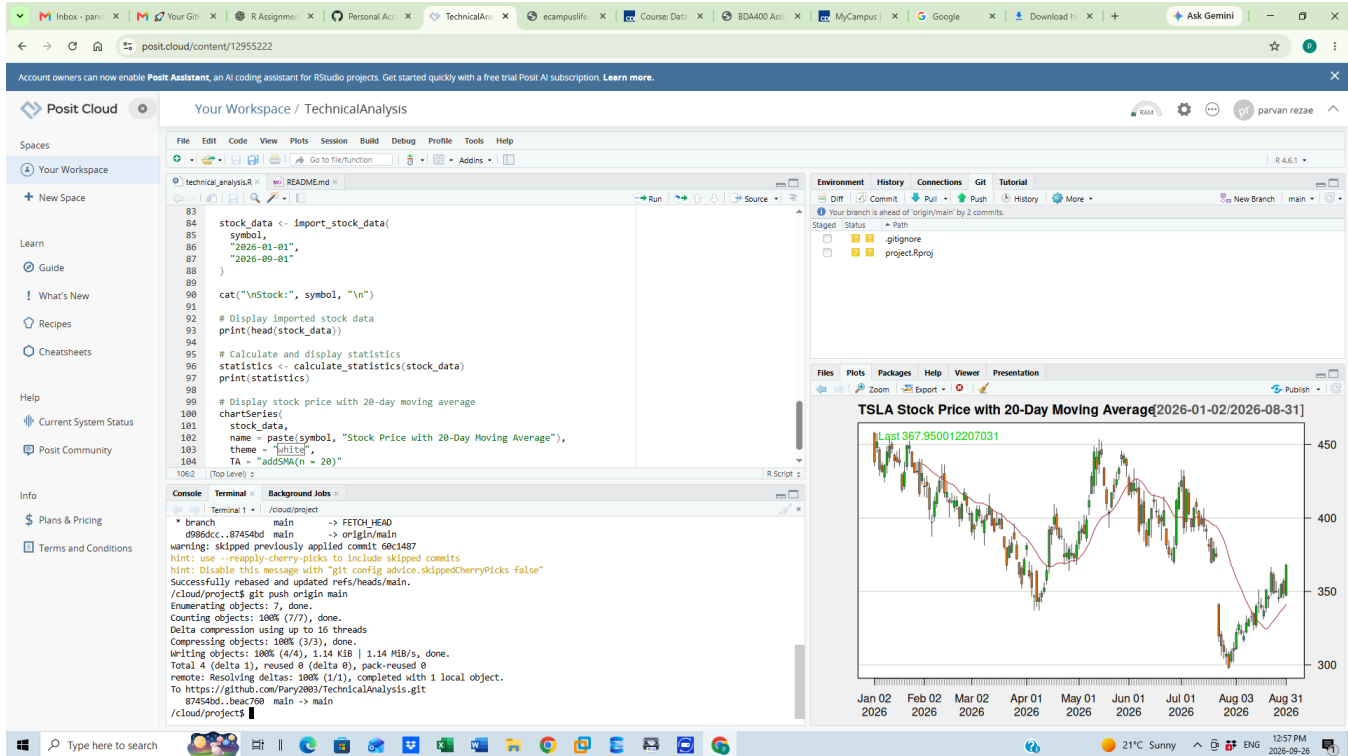


Figure 7. Successful Git push from Posit Cloud to the GitHub main branch.

7. Final GitHub Verification

The final GitHub version of technical_analysis.R contains 106 lines and shows the completed commit, confirming that the updated assignment code is available in the repository.

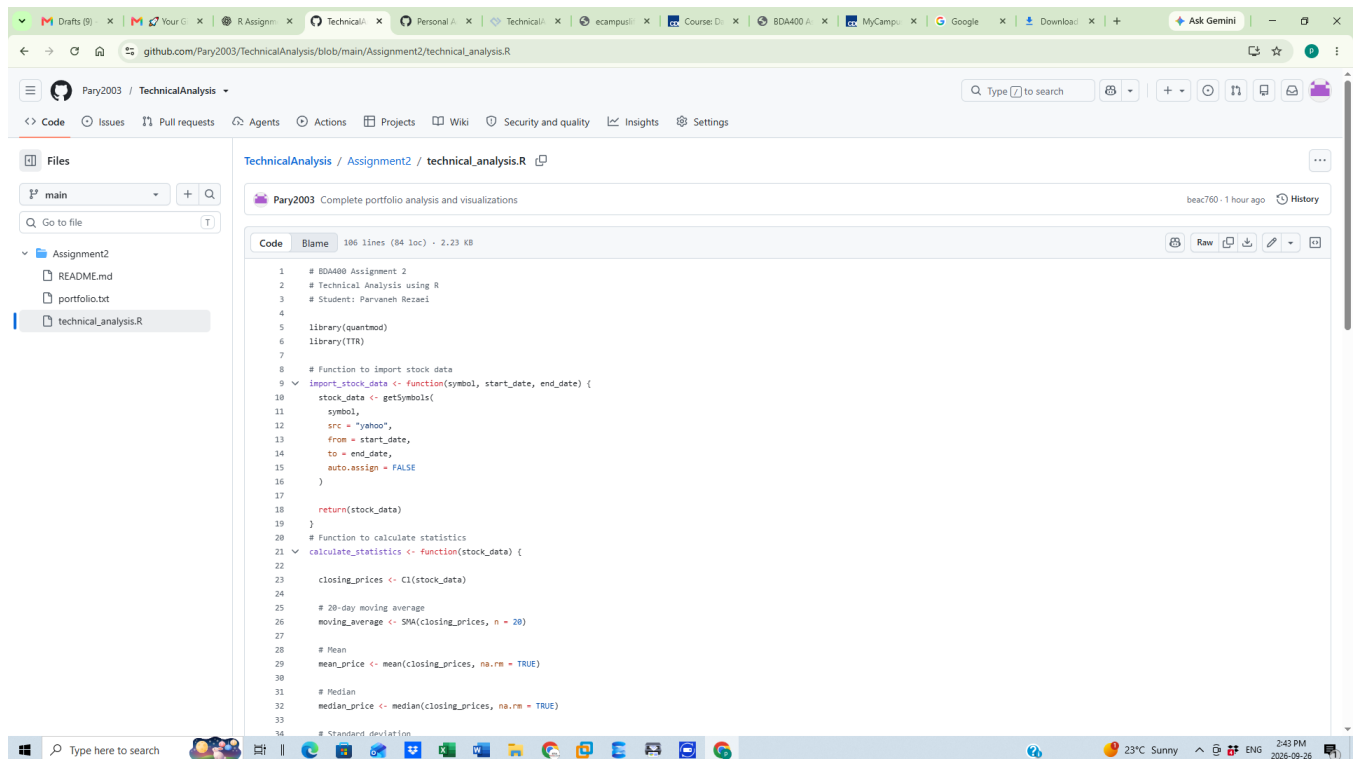


Figure 8. Final `technical_analysis.R` file on GitHub after the successful push.

8. Conclusion

The preliminary technical-analysis project was completed in R using Posit Cloud, Git, and GitHub. The project imports stock data, calculates descriptive statistics and a 20-day moving average, processes a five-stock portfolio, creates visualizations, and stores the completed work in the GitHub repository.